

WASP-2b

R-1503

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_210702 · created 2026-08-01T15:04:32+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459397.9085 +/- 0.0066 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.046
Transit depth (Rp/Rs)^2	2.33 +/- 1.41 [%]
Orbital Inclination (inc)	84.51 +/- 2.88
Airmass coefficient 1 (a1)	0.29 +/- 0.14
Airmass coefficient 2 (a2)	1.09 +/- 1.15
Scatter in the residuals of the lightcurve fit is	49.33 %
Best Comparison Star	#2 - [431, 225]
Optimal Method	PSF photometry
Transit Duration (day)	0.062 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.153 ± 0.046 vs 0.1326 (deviation 0.44σ)
Duration fit vs published	0.062 d vs 0.0737 d (deviation 0.35σ)
Mid-time O–C	16.6 min
Depth SNR	3.3
Residual scatter	49.33 %

Light curve

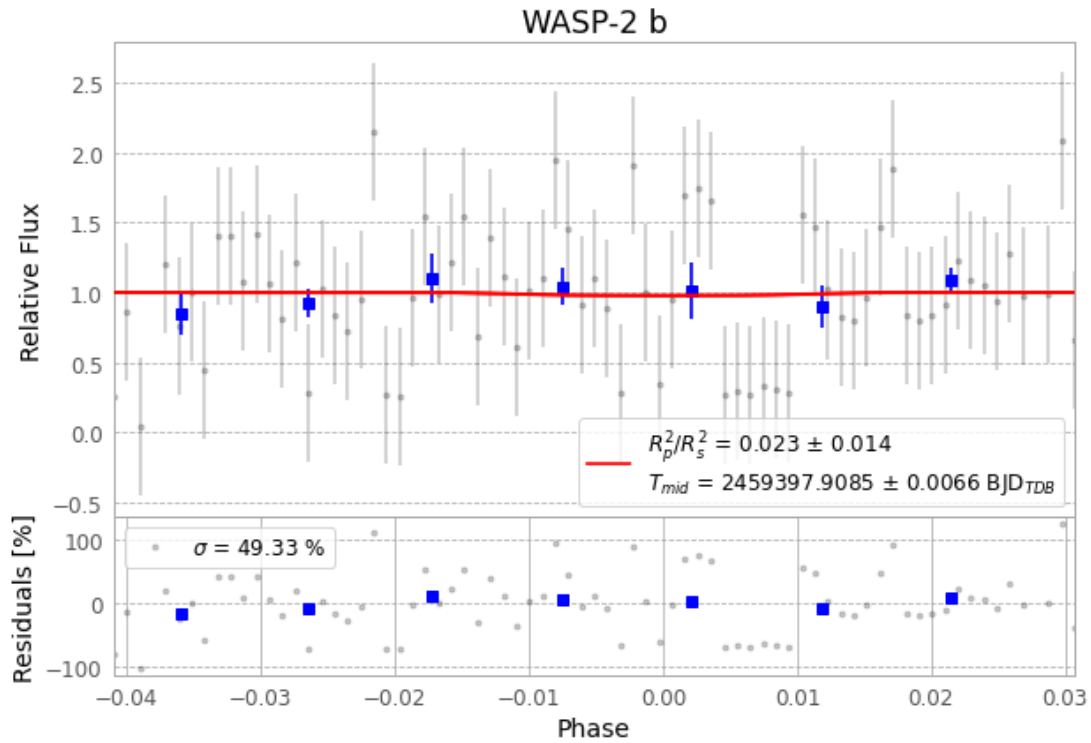


Figure 1 — FinalLightCurve_WASP-2 b_2021-07-02.png (SHA-256 ee904ff5ba4c7599...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [387, 107], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bb2ca0d3097e8672...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-2210702073312.FITS → WASP-2210702111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-210702.log (9317b96e18c4...), FinalLightCurve_WASP-2 b_2021-07-02.png (ee904ff5ba4c...), FinalLightCurve_WASP-2 b_2021-07-02.csv (69dd85a17431...)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 15:04Z.